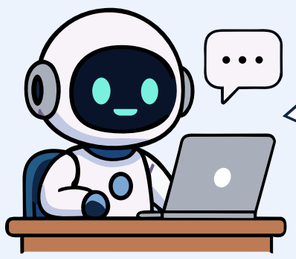
Training-Free Framework for Single-Turn Web Intent Understanding with Retrieval-Augmented Reasoning and Calibrated Refusal

INPUT: a single-turn user query q (context-sparse, web search / IM)



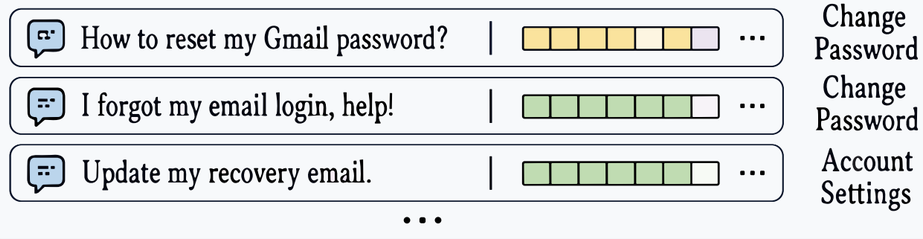
q (example)
"How do I change my email password on Gmail?"

External Intent Knowledge Base (KB, zero-annotation, updatable at inference)

Intent Prototypes (centroid embeddings)

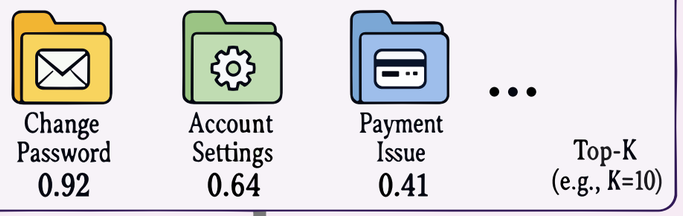


Utterance Vectors (*examples per intent*)



Module 1: Retrieval & Reasoning

1) Top-K intent retrieval from KB



2) LLM self-assessed reasoning & confidence



Candidate intents (by rank)	$Conf_{LLM}$
1. Change Password (primary)	0.86
2. Account Settings (secondary)	0.42
3. Payment Issue	0.17
...	

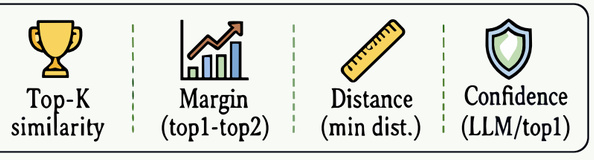
Outputs: candidate intents with self-assessed confidence $Conf_{LLM}$ (primary, secondary)

Module 2: Set-Valued Conformal Calibration

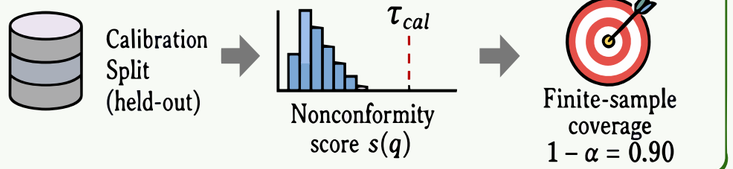
1) Rank-normalized score fusion

$$s(q) = g(stats_{ret}, Conf_{LLM})$$

$stats_{ret}$ =



2) Split conformal calibration (no gradients)

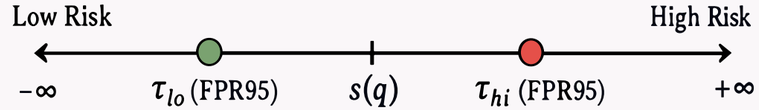


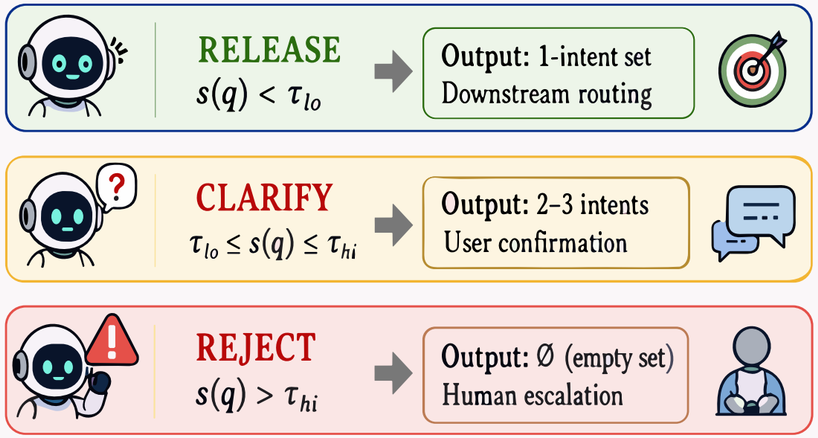
3) Theoretical guarantees

- Marginal coverage: $Pr(Y \in \hat{S}(q)) \geq 1 - \alpha = 0.90$
- Set-level Jaccard risk $\leq \varepsilon_J$ (finite-sample bound)
- No gradient training, no classifier head

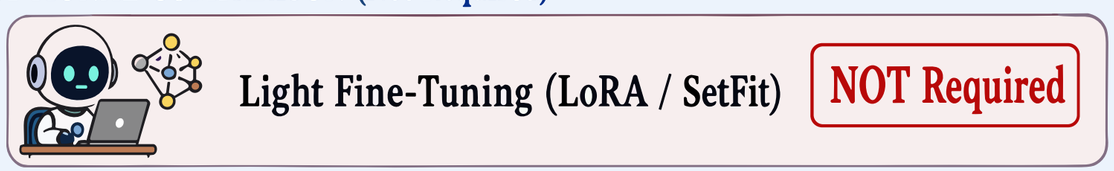
Module 3: Adaptive Three-State Decision

Thresholds τ_{lo}, τ_{hi} calibrated at FPR95 on $s(q)$





OPTIONAL COMPARISON (Not Required)



Retrieval + Calibration + Decision Only with Theoretical Guarantees